

Nicoló La Porta

PHD STUDENT · INFORMATICS

*Institute of Information System and Networking (SUPSI)
Faculty of Informatics, University of Lugano (USI)*

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Education

University of Lugano

PHD IN INFORMATICS

Lugano, Switzerland

September 2023 - September 2027

- Academic Advisor: Professor Dr. Marc Langheinrich
- Research Advisor: Dr. Michela Papandrea

Polytechnic University of Turin

MASTER OF SCIENCE IN BIOMEDICAL ENGINEERING *cum laude*

Turin, Italy

September 2019 - July 2022

- Thesis advisor: Full Professor Filippo Molinari
- External Advisor: Eng. Alessandro Puiatti
- Thesis: Sleep apnea events recognition based on polysomnographic signals: a machine learning approach

University of Palermo

BACHELOR OF SCIENCE IN BIOMEDICAL ENGINEERING *cum laude*

Palermo, Italy

September 2016 - July 2019

- Thesis advisor: Professor Massimiliano Zingales
- Thesis: Biomechanical characterization of sutures in porcine menisci

Professional Experience

- 2023-ongoing **PhD Researcher**, Institute of Information System and Networking (ISIN), University of Applied Sciences and Arts of Southern Switzerland (SUPSI) and Faculty of Informatics, University of Lugano (USI)
- 2022-2023 **Research Intern**, Institute of Digital Technologies for Personalised Healthcare (MeDiTech), University of Applied Sciences and Arts of Southern Switzerland (SUPSI)

Publications

PUBLISHED

- La Porta, L.**, Di Maio, C., Fraci, F.D., Glaser, B., Ramelli, G.P. & Papandrea, P. (2026). Wear2Toy: Leveraging Wearable Devices for Hand-Held Objects Activity Recognition. *In Proceedings of the 2026 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom)*. [doi](#)
- Fiorillo, L., Lombardi, G., **La Porta, N.**, Arnaud, L., Veneruso, M., Castelnovo, A., Bertaina, I., Staedler, C., Kaelin-Lang, A. & Galati, S. (2025). Altered wakeful theta activity characterizes levodopa-induced dyskinesia in Parkinson's disease. *npj Parkinsons Dis.* 12, 115 (2026). [doi](#)
- Monachino, G., **La Porta, N.**, Zanchi, B., Fiorillo, L., Dei Rossi, A., Farino, G. & Faraci, F. D. (2025). Self-DANA: A Resource-Efficient Channel-Adaptive Self-Supervised Approach for ECG Foundation Models. Presented at *TS4H workshop - NeurIPS 2025*. [doi](#)
- Gelsomini, M., **La Porta, N.**, Fusco, F., & Papandrea, M. (2025). Designing for Attention: Evaluating an Affordable Immersive VR Platform for Neurodiverse Children. *In Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility* (pp. 1-18). [doi](#)
- Centorame, L., **La Porta, N.**, Papandrea, M., Mancini, A., & Foppa Pedretti, E. (2025). Evaluation of Feature Selection and Regression Models to Predict Biomass of Sweet Basil by Using Drone and Satellite Imagery. *Applied Sciences*. 15(11):6227., 6(1). [doi](#)
- La Porta, N.**, Oldano, G., Puiatti, A., Leidi, T., & Papandrea, M. (2025). Toward unbiased emotion recognition: overcoming user bias with siamese convolutional networks. *Signal, Image and Video Processing*, 19(11), 875. [doi](#)
- La Porta, N.**, Minardi, L., & Papandrea, M. (2024). Fine-Grained Human Activity Recognition Through Dead-Reckoning and

Temporal Convolutional Networks. *In International Workshop on Sensor-Based Activity Recognition and Artificial Intelligence (pp. 3-17). Cham: Springer Nature Switzerland* [doi](#)

La Porta, N., Scafa, S., Papandrea, M., Molinari, F., & Puiatti, A. (2024). Sleep apnea events recognition based on polysomnographic recordings: a large-scale multi-channel machine learning approach. *IEEE open journal of engineering in medicine and biology*. [doi](#)

Nocilli, M., Scafa, S., **La Porta, N.**, Ghislieri, M., Agostini, V., Moraud, E. M., & Puiatti, A. (2024). GAIT: gait analysis interactive tool a pipeline for automatic detection of gait events across different motor impairments. *Signal, Image and Video Processing*, 18(12), 8499-8506. [doi](#)

Papandrea, M., Peternier, A., Frei, D., **La Porta, N.**, Gelsomini, M., Allegri, D., & Leidi, T. (2024). V-Cockpit: A Platform for the Design, Testing, and Validation of Car Infotainment Systems through Virtual Reality. *Applied Sciences*, 14(18), 8160. [doi](#)

IN REVIEW

Monachino, G., **La Porta, N.**, Zanchi, B., Fiorillo, L., Dei Rossi, A., Farina, G. & Faraci, F.D. (2026). Self-DANA: A Resource-Efficient Channel-Adaptive Self-Supervised Approach for ECG Foundation Models. *Elsevier - Computers in Biology and Medicine*.

Awards, Fellowships, & Grants

21.05.2026	Innovation Booster Grant - Future Food Farming , InnoSuisse	CHF 39'000,00
	Best Paper Award , 22nd edition of the workshop on Context and Activity Modeling and Recognition (CoMoRe-AI) collocated with the 24th Annual IEEE Intl. Conference on Pervasive Computing and Communications (PerCom'26)	
02.03.2026	Travel Grant , PerCom'26 travel grant selection committee	EUR 250,00

Teaching Assistance

Spring 2025	Multimodal Recommendation Systems and Complex Networks , MSE Data Science	SUPSI
Fall 2025	Supervised Learning , B.Sc. Data Science	SUPSI
Spring 2024	Multimodal Recommendation Systems and Complex Networks , MSE Data Science	SUPSI
Fall 2024	Supervised Learning , B.Sc. Data Science	SUPSI
Spring 2024	Unsupervised Learning , B.Sc. Data Science	SUPSI
Fall 2023	Supervised Learning , B.Sc. Data Science	SUPSI

Mentoring

BACHELOR STUDENTS

2026	Gabriele Boni , B.Sc. Computer Science, Thesis, <i>TRACES – TRacking and Aligning Camera Entities and Sensors data</i>	SUPSI
2026	Fabio Ceccatelli , B.Sc. Computer Science, Semester Project, <i>TRACE – Tracking and Recognition of Active Camera Entities</i>	SUPSI
2026	Gabriele Boni , B.Sc. Computer Science, Semester Project, <i>TRACE – Tracking and Recognition of Active Camera Entities</i>	SUPSI

MASTER STUDENTS

2025	Andrea Cosentino , MSE Data Science, Semester Project, <i>Predicting ADOS-2 scores from hand-held object manipulation sequences.</i>	SUPSI
2026	Carolina Di Maio , MSE Computer Science, Thesis, <i>Exploring Self-supervised Learning for Play Activity Recognition in Children affected by Autism Spectrum Disorder</i>	SUPSI
2025	Carolina Di Maio , MSE Computer Science, Semester Project, <i>Exploring Transfer Learning for Play Activity Recognition in Children affected by Autism Spectrum Disorder</i>	SUPSI
2025	Andrea Cosentino , MSE Data Science, Semester Project, <i>v-CockPit: inferring drivers' arousal from PPG signal during VR driving simulations</i>	SUPSI
2024	Mattia Bosetti , MSE Data Science, Semester Project, <i>v-CockPit: recognizing drivers' emotions during VR driving simulations</i>	SUPSI